

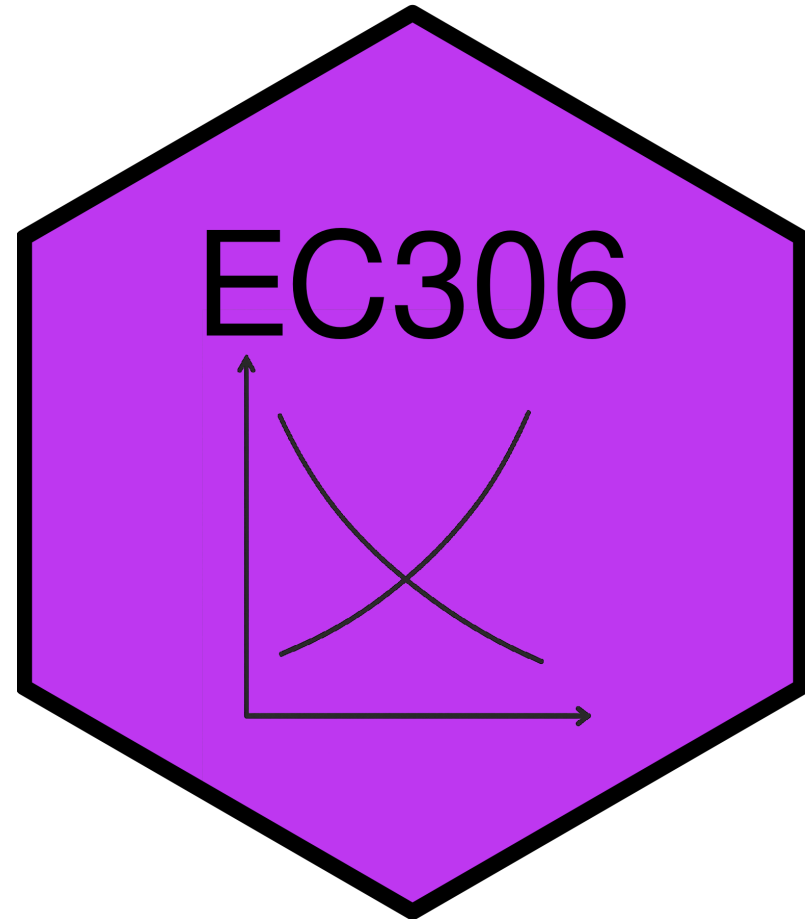
Labour Demand

EC306- Wages and Employment

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Goals of This Section

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In this section we will:

- Understand how firms decide how much labour to employ
- Compare short-run vs. long-run labour demand decisions
- Explain why labour demand is downward sloping
- Learn factors that affect the elasticity of labour demand
- Distinguish between workers and hours as an input into production
- Add costs that are independent of hours worked (quasi-fixed costs)

Background

Background

Labour demand: the number of workers firms want to hire at a given wage rate

! Derived Demand

Labour demand is linked to demand for goods/services produced — it is an input into that production

Several factors influence labour demand:

- Firm's objectives and constraints
- Demand conditions in product markets
- State of technology
- Time horizon
 - Short run: one or more inputs fixed (usually capital)
 - Long run: all inputs variable

Structure of Product Markets

- Labour demand is driven by the firm's output (labour is an input)
- Structure of the product market influences demand for labour
 - Ranges from perfect competition (many sellers) to monopoly (one seller)
- Structure of the labour market influences supply the firm faces
 - Ranges from perfect competition (many workers) to monopsony (one buyer)

Note

In this section we examine **perfect competition** in both product and labour markets

Demand in the Short Run

Model

As labour demand is derived from production, the model starts with a production function:

$$Q = F(K, N)$$

Where:

- Q is quantity of output
- K is capital (fixed in short run at K_0)
- N is labour (variable in short run)

Firm's objective: maximize profit

In the short run this means choosing N given fixed K_0

Model

To see this mathematically:

Output prices are given (perfect competition)

Profit:

$$\begin{aligned}\pi(N) &= pQ - wN - rK_0 \\ &= pF(K_0, N) - wN - rK_0\end{aligned}$$

Differentiate $\pi(N)$ w.r.t. N :

$$\frac{d\pi}{dN} = p \frac{\partial F}{\partial N}(K_0, N) - w = pMP_N - w$$

Model

Optimality Condition

Optimal N^* satisfies: $pMP_N = w$

Where:

- MP_N is the **marginal product of labour**
- pMP_N is the **marginal revenue product of labour** (MRP)
 - When firm is in perfectly competitive product market and $MR = p$ it is called **value of marginal product** (VMP)
- w is the wage rate (marginal cost of labour, MC_N)

Optimality requires that the value of the last unit of labour hired equals its cost

Model

Other key metrics:

- Total Revenue Product (TRP): Total revenue associated with labour employed

$$TRP = p \times Q$$

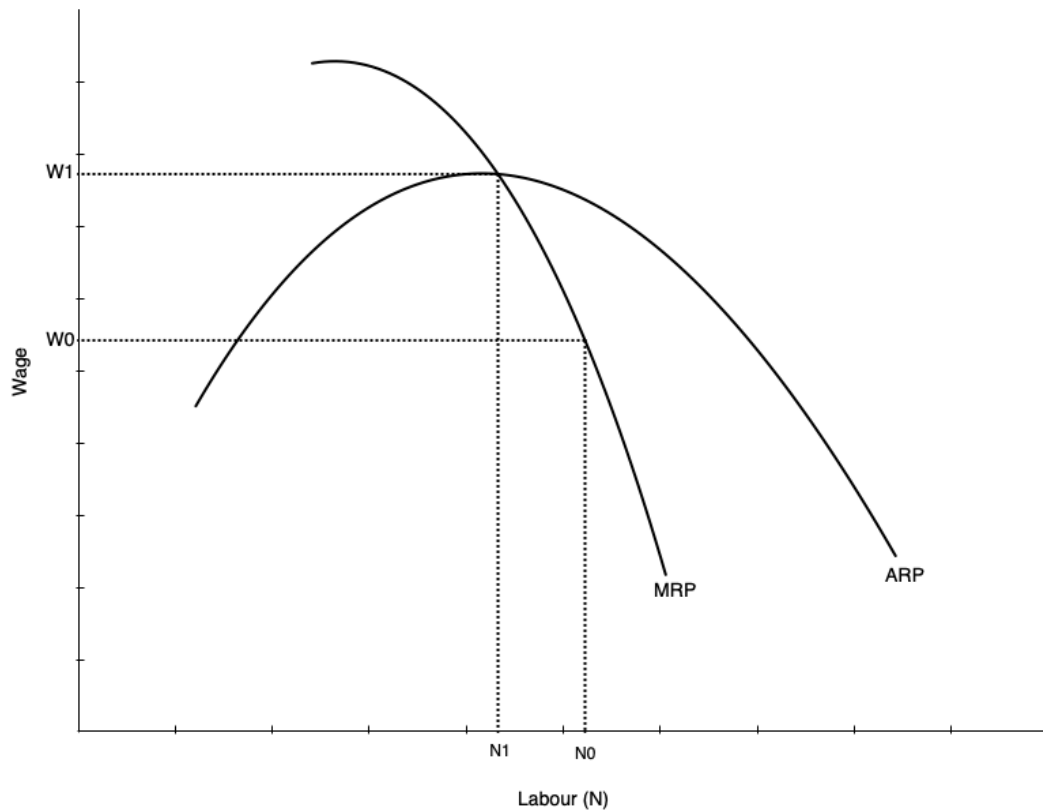
- Average Revenue Product (ARP): Average revenue per unit of labour employed

$$ARP = \frac{TRP}{N} = \frac{p \times Q}{N}$$

The firm's optimal choice of labour is where $MRP_N = w$

- This defines its labour demand curve in the short run
- As w varies, the firm adjusts N to maintain equality

Labour Demand Curve in Short Run



The short run labour demand curve is the MRP curve below ARP

- Shows amount of labour demanded at every wage
- Slopes down because of **diminishing marginal product** of labour
 - As more labour is employed, extra output produced falls
 - Not because of inferior labour, but because more labour uses fixed capital

As wage falls, firm is willing to use more labour

Demand Curve and Competition in Product Market

- Labour demand depends on demand for the product being produced
- Also depends on structure of the product market
- Firm always maximizes profit by choosing labour where $MRP_N = w$

But marginal revenue differs across market structures:

- In **perfect competition**: $MR = p$, and MR does not fall with more N
- In **monopoly**: $MR < p$, and MR falls as N increases

Warning

With less competition in product market, labour demand is lower and falls faster with more N

Think About It

A firm currently employs workers where $pMP_N = w$. Suppose the government introduces a per-unit subsidy on the firm's output, raising the effective price the firm receives. What happens to the firm's short-run demand for labour, and why?

Demand in the Long Run

Introduction

Long Run Differs from Short Run

All inputs are variable — firms can adjust capital as well as labour

This makes the decision more complicated:

- Must consider substitution between labour and capital
- Must consider economies of scale

Approach to find optimum occurs in two steps:

1. Minimize cost holding output constant
2. Choose output to maximize profit given cost function

Cost Minimization

First step: minimize cost of producing given output Q_0

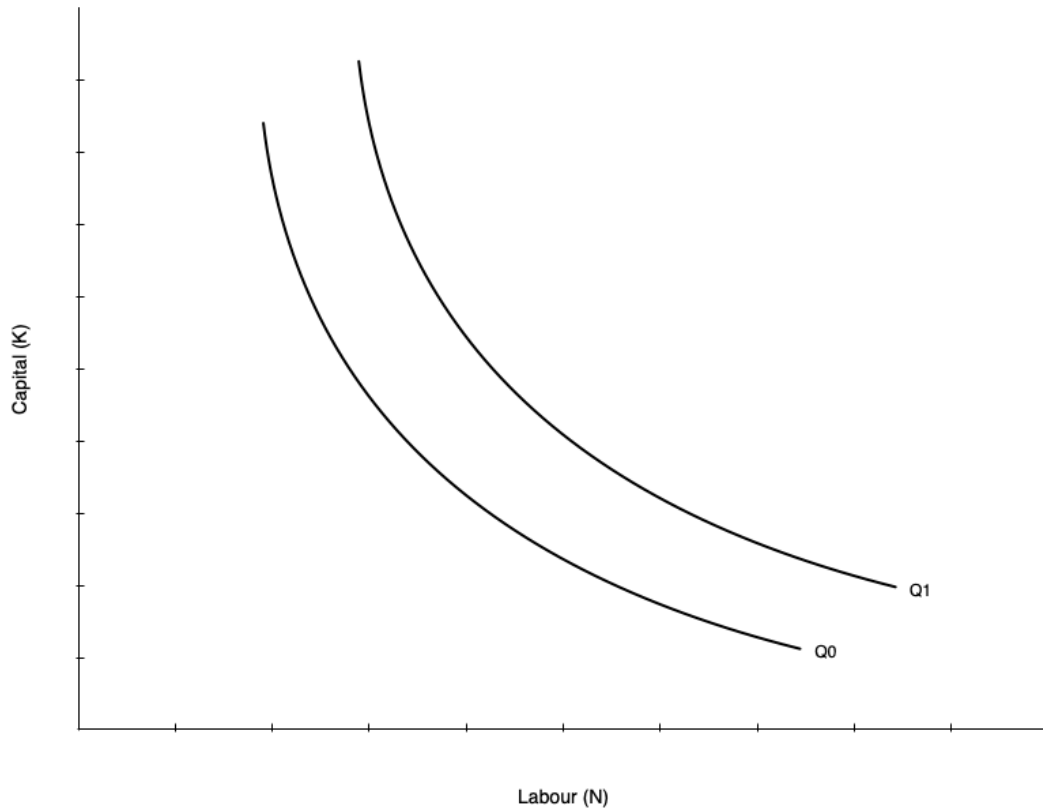
Occurs when an isoquant is tangent to an isocost line:

- **Isoquant**: shows combinations of K and N that produce Q_0
- **Isocost line**: shows combinations of K and N that cost the same amount
- Analogous to utility maximization with indifference curves and budget constraints

Assume cost of capital is r and cost of labour is w

- Total cost: $C = rK + wN$
- Production function: $Q = F(K, N)$
- Minimize C subject to $Q = Q_0$

Cost Minimization



Graph shows two isoquants

- Combinations of capital and labour that produce the same quantity
- Shape given by the production function
 - Shown as convex, but could be different shapes
 - Linear isoquant → perfect substitutes
 - L-shaped isoquant → perfect complements

Slope of isoquant is the MRTS

- Marginal rate of technical substitution
- Rate at which you can substitute labour for capital holding output constant

Cost Minimization

The slope of the isoquant is given by the MRTS:

$$MRTS = \frac{MP_N}{MP_K}$$

Tells you amount of capital you can give up for one more unit of labour holding output constant

Diminishing MRTS

Isoquants typically display diminishing MRTS — when a lot of capital is used, you can give up a lot for one more unit of labour; when a lot of labour is used, you can give up only a little capital

Cost Minimization

The isocost line gives combinations of K and N that cost the same amount

Total cost:

$$C = rK + wN$$

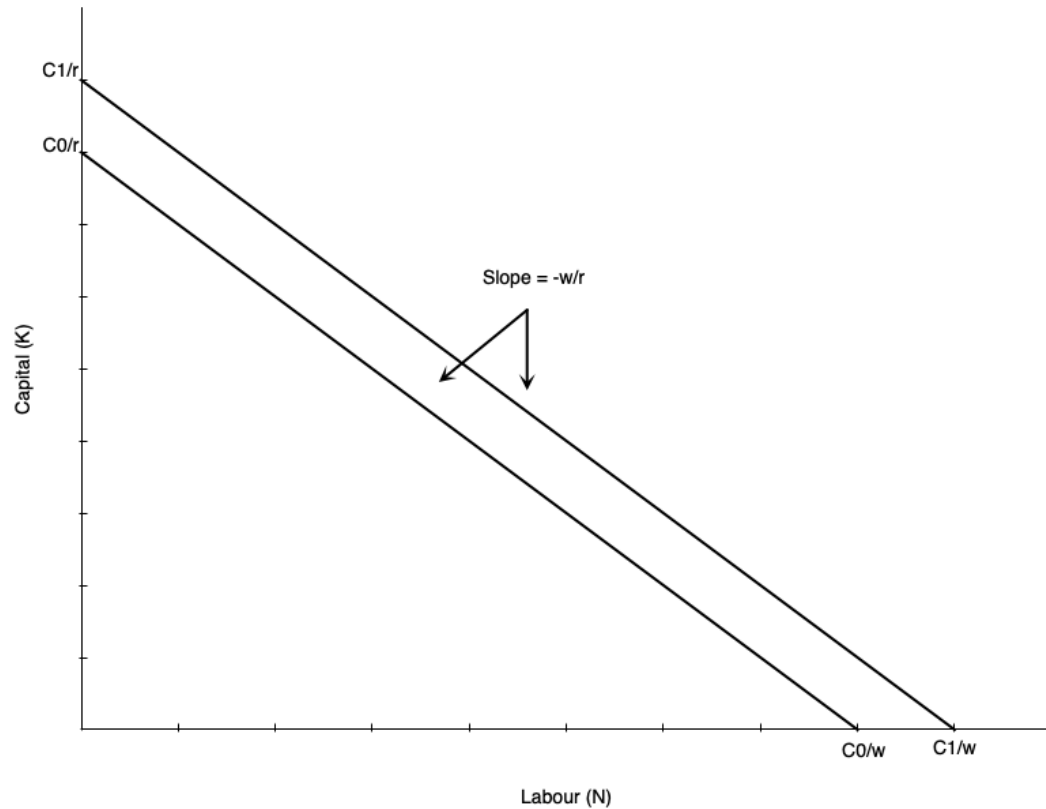
Rearranging gives the isocost line equation:

$$K = \frac{C}{r} - \frac{w}{r}N$$

Slope of isocost line: ratio of input prices

- As w increases, isocost line becomes steeper
- As r increases, isocost line becomes flatter

Cost Minimization

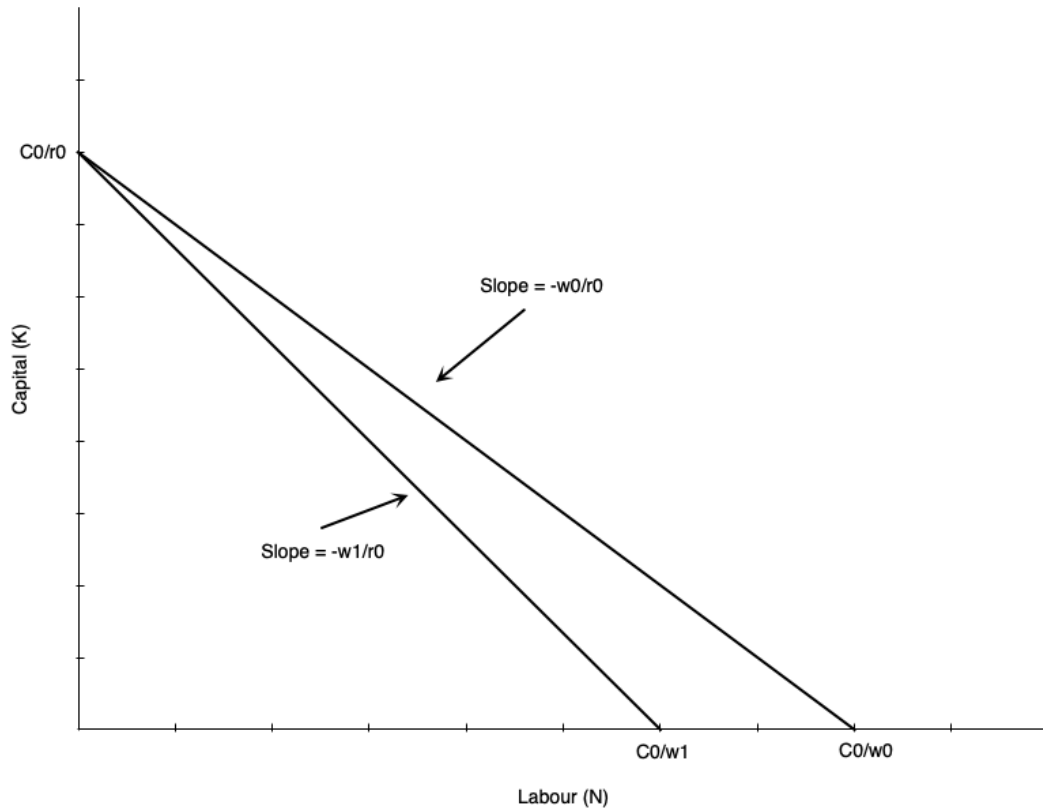


Example isocost:

- Slope is ratio of input prices $-w/r$
- If firm uses only labour: C/w
- If firm uses only capital: C/r
- As labour falls by one unit, capital must rise by w/r units to keep cost constant

Higher cost line shifts isocost outward (when C_0 increases to C_1)

Cost Minimization



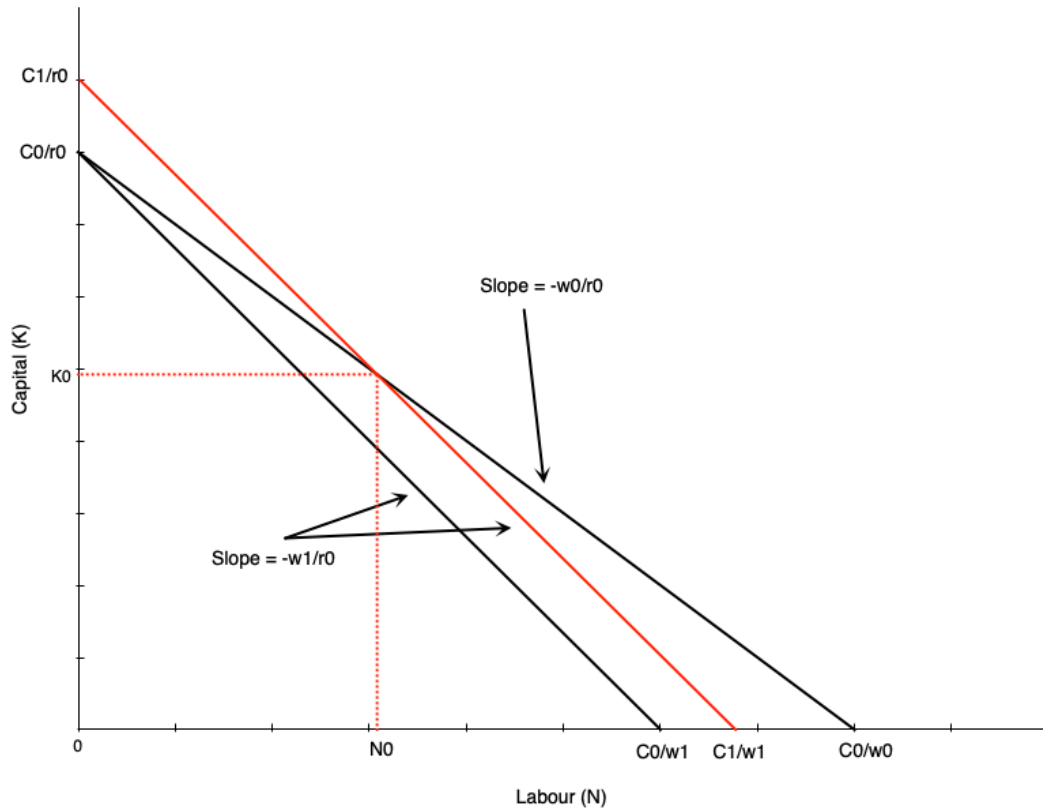
Suppose wage rises from w_0 to w_1

Holding cost constant at C_0 :

- Isocost line pivots inward
- Steeper slope as w/r increases
- If firm uses only capital, vertical intercept stays at C_0/r
- If firm uses only labour, horizontal intercept falls to C_0/w_1

All combinations of new isocost lie below old isocost

Cost Minimization



To use any of the old combinations of K and N firm must increase cost

- Shifts isocost outward to C_1
- Magnitude of shift depends on which combination is chosen

Example depicted shows cost curve shifting out to use combination (N_0, K_0) at new prices

Cost Minimization

Cost Minimizing Bundle

Where isoquant is tangent to isocost line

In mathematical terms:

$$\frac{MP_N}{MP_K} = \frac{w}{r}$$

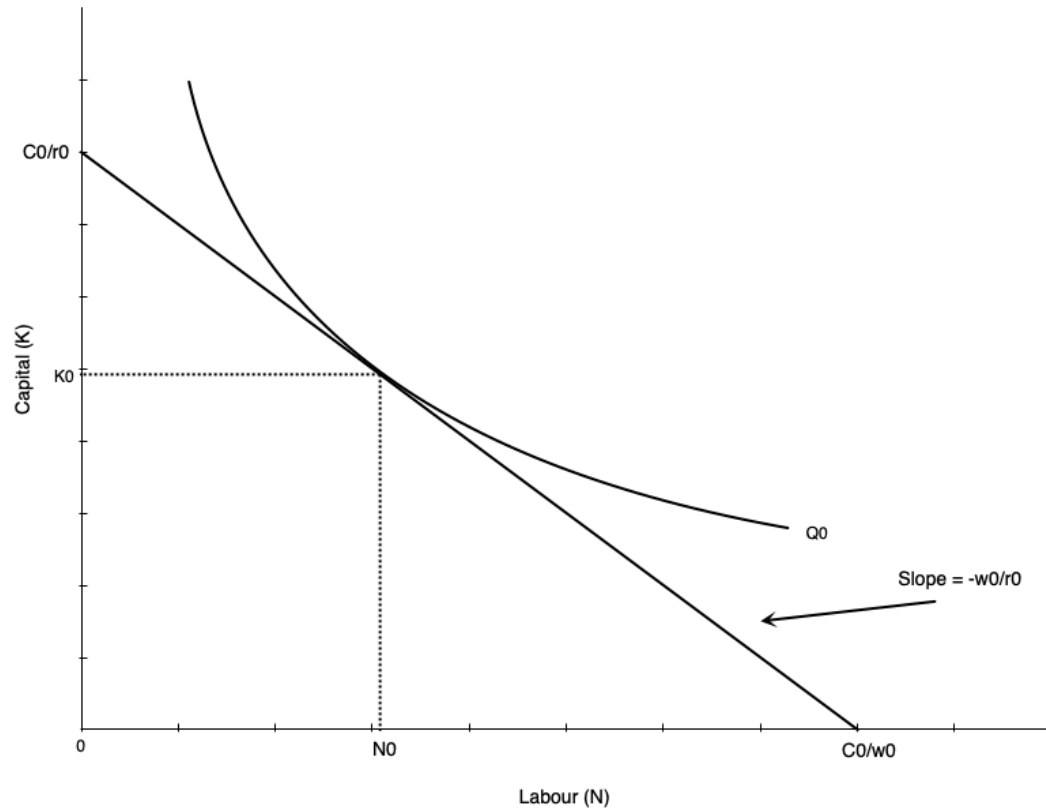
Says that MRTS equals the ratio of input prices

Alternatively:

$$\frac{MP_N}{w} = \frac{MP_K}{r}$$

The last dollar spent on each input must yield the same marginal product

Cost Minimization



Cost minimization depicted

- Hold output constant at Q_0
- Find optimal combination of K and N
- Happens at tangency point between isoquant and isocost line

Cost Minimization

Two-Step Process

We have only done cost minimization here — lowest cost given output level Q_0 . Profit maximization involves choosing output level as well.

- Second step: choose Q to maximize profit given cost function
- Second step is not depicted here

If wage rate goes up, firm re-optimizes:

- Firm marginal and average cost curves shift up
- Each firm reduces output
- Market price rises
- Isoquants at profit max shift inward

Cost Minimization – Math

Suppose production exhibits **decreasing returns to scale**:

$$y = F(K, N) = K^{\alpha} N^{\beta}, \quad \alpha > 0, \beta > 0, \alpha + \beta < 1$$

Input prices: r (capital), w (labour)

Cost-minimization problem:

$$\min_{K, N} C = rK + wN \quad \text{s.t.} \quad K^{\alpha} N^{\beta} \geq y$$

Lagrangian:

$$\square = rK + wN + \lambda(y - K^{\alpha} N^{\beta})$$

Cost Minimization – Math

First-order conditions:

$$\frac{\partial \square}{\partial K} = r - \lambda \alpha K^{\alpha-1} N^{\beta} = 0$$

$$\frac{\partial \square}{\partial N} = w - \lambda \beta K^{\alpha} N^{\beta-1} = 0$$

$$y = K^{\alpha} N^{\beta}$$

Divide the first two conditions to eliminate λ :

$$\frac{\beta}{\alpha} \frac{K}{N} = \frac{w}{r}$$

This is the familiar result: $\frac{MP_N}{MP_K} = \frac{w}{r}$

Cost Minimization – Math

From production $y = K^\alpha N^\beta$:

$$N = \left(\frac{y}{K^\alpha} \right)^{\frac{1}{\beta}}$$

Then:

$$\frac{K}{N} = K \left(\frac{K^\alpha}{y} \right)^{1/\beta} = K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta}$$

Cost Minimization – Math

Plug into $\frac{\beta}{\alpha} \frac{K}{N} = \frac{w}{r}$:

$$\frac{\beta}{\alpha} K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta} = \frac{w}{r}$$

$$K^{\frac{\alpha+\beta}{\beta}} = \frac{\alpha}{\beta} \frac{w}{r} y^{1/\beta}$$

Solving for optimal K :

$$K^*(y; w, r) = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha w}{\beta r} \right)^{\frac{\beta}{\alpha+\beta}}$$

Cost Minimization – Math

Use $\frac{K}{N} = \frac{\alpha w}{\beta r}$ to get N^* :

$$N^*(y; w, r) = \frac{\beta r}{\alpha w} K^* = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha w}{\beta r} \right)^{-\frac{\alpha}{\alpha+\beta}}$$

The cost function:

$$C(w, r, y) = rK^*(y; w, r) + wN^*(y; w, r) = \phi(w, r) y^{\frac{1}{\alpha+\beta}}$$

Cost Minimization – Math

Where:

$$\phi(w, r) = r \left(\frac{\alpha w}{\beta r} \right)^{\frac{\beta}{\alpha+\beta}} + w \left(\frac{\alpha w}{\beta r} \right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

Stage 2: choosing output level to maximize profit

Suppose firm is a price taker in product market with price p

Profit:

$$\pi(y) = py - C(y; w, r) = py - \phi(w, r) y^{\frac{1}{\alpha+\beta}}$$

Optimize profit with respect to y :

$$\frac{d\pi}{dy} = p - \phi(w, r) \cdot \frac{1}{\alpha + \beta} \cdot y^{\frac{1}{\alpha+\beta}-1}$$

Optimal output level (set $\frac{d\pi}{dy} = 0$):

$$y^*(w, r, p) = \left(\frac{(\alpha + \beta) p}{\phi(w, r)} \right)^{\frac{\alpha+\beta}{1-(\alpha+\beta)}}$$

Cost Minimization – Math

You can then plug in to get optimal inputs:

$$K^*(w, r, p) = \left(\frac{(\alpha + \beta) p}{\phi(w, r)} \right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha w}{\beta r} \right)^{\frac{\beta}{\alpha+\beta}}$$

$$N^*(w, r, p) = \left(\frac{(\alpha + \beta) p}{\phi(w, r)} \right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha w}{\beta r} \right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

 Don't Get Bogged Down

I will not ask you to recreate this derivation on exams

Key points:

- First minimize cost → optimal inputs as function of output and input prices
 - As you vary any of those three things, the inputs change
- Given cost function, choose output to maximize profit
 - Output depends on product price and input prices
 - Changes in input prices change both optimal inputs and output level

Think About It

A firm currently satisfies $\frac{MP_N}{w} = \frac{MP_K}{r}$. Suppose the rental rate of capital r suddenly doubles while the wage w stays the same. In what direction will the firm adjust its mix of labour and capital, and what happens to the isocost line?

Deriving Labour Demand in Long Run

Deriving Labour Demand

The labour demand curve plots combinations of w and N such that firm is maximizing profit

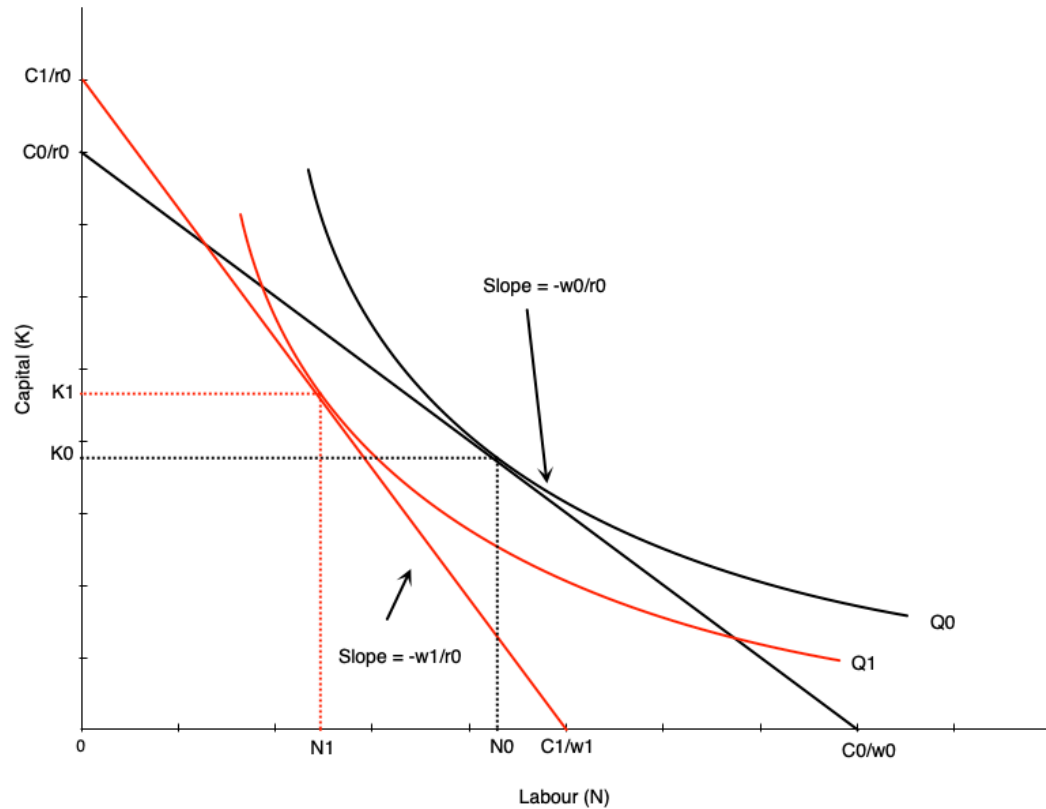
To derive it, consider changing wage from w_0 to w_1 :

- Isocosts get steeper as w/r increases
- Firm finds cost minimizing combination of K and N at new wage for each output level
- Because cost increases, firm reduces output (optimal isoquant shifts left)

At new optimum:

- Lower quantity of labour demanded
- Higher wage

Deriving Labour Demand

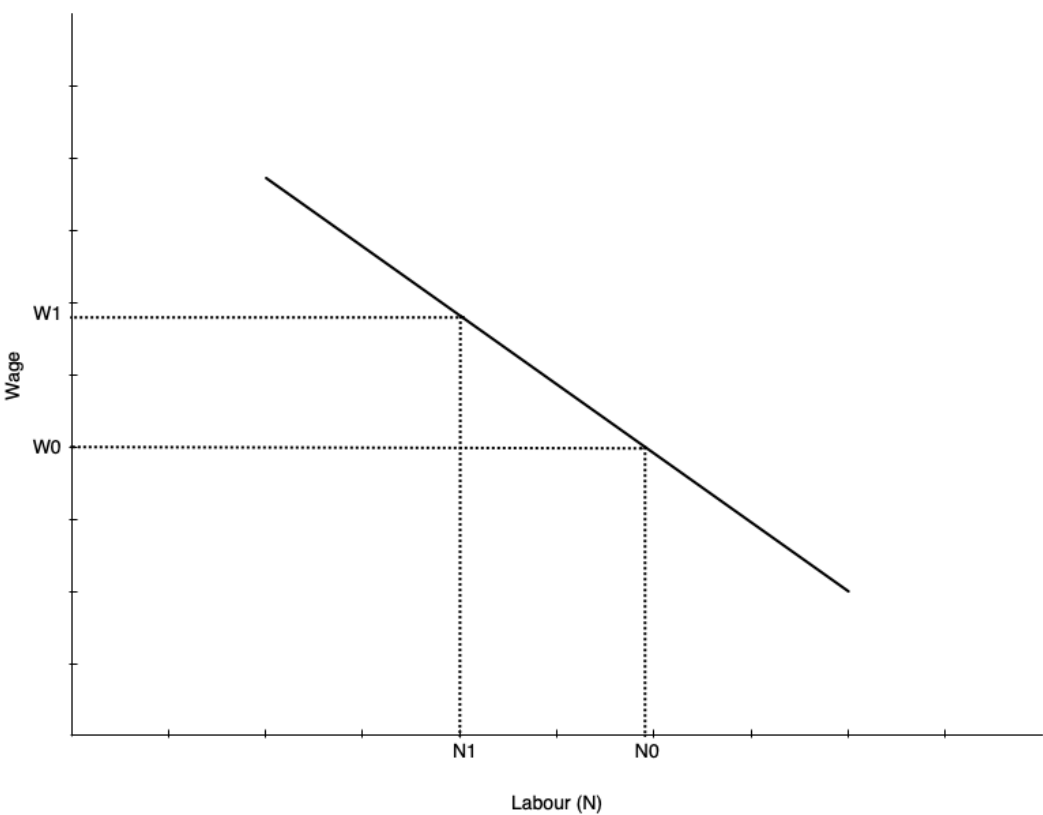
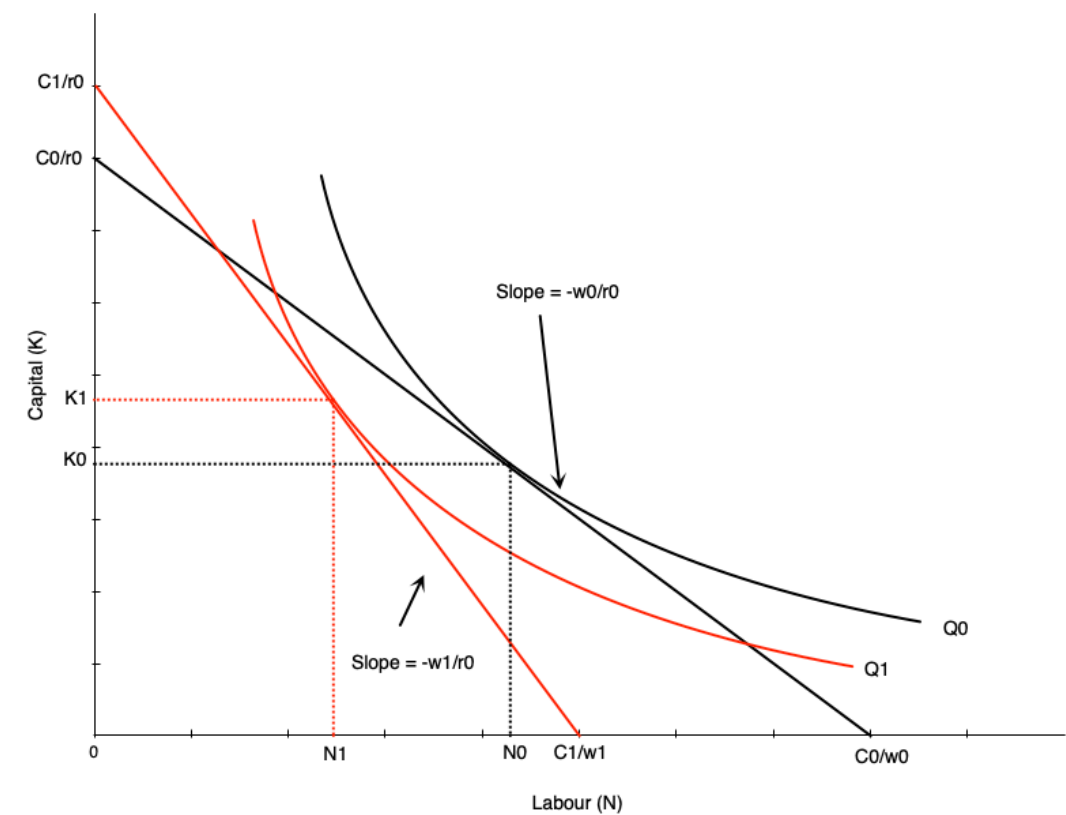


Effect of wage increase on labour:

- Wage goes up $\rightarrow$ isocost line steepens
- Firm scales down production to lower isoquant
- Tangency between new isoquant and new isocost shows new optimal labour and capital
- This involves less labour employed

Effect on capital: depends on substitutability between labour and capital (shape of isoquant)

Deriving Labour Demand



Deriving Labour Demand

The labour demand curve summarizes optimal choices of labour at different wage rates

- Previous slide shows two points on labour demand curve: (w_0, N_0) and (w_1, N_1)
- To derive full demand curve, vary wage rate to other levels

⚠ Ceteris Paribus

Demand curve shows amount of labour demanded at every wage **all else equal** — other determinants will shift the curve

Scale and Substitution Effects

Two Effects When Wage Changes

- **Substitution effect:** change in input mix due to change in relative input prices
- **Scale effect:** change in output due to change in cost of production

Substitution effect:

- Makes firms substitute capital for labour
- Leads to less labour demanded when wage rises (more capital demanded)

Scale effect:

- Comes from reduction in output
- As production becomes more expensive, firms produce less
- Leads to less labour demanded when wage rises (use less labour and capital)

Scale and Substitution Effects

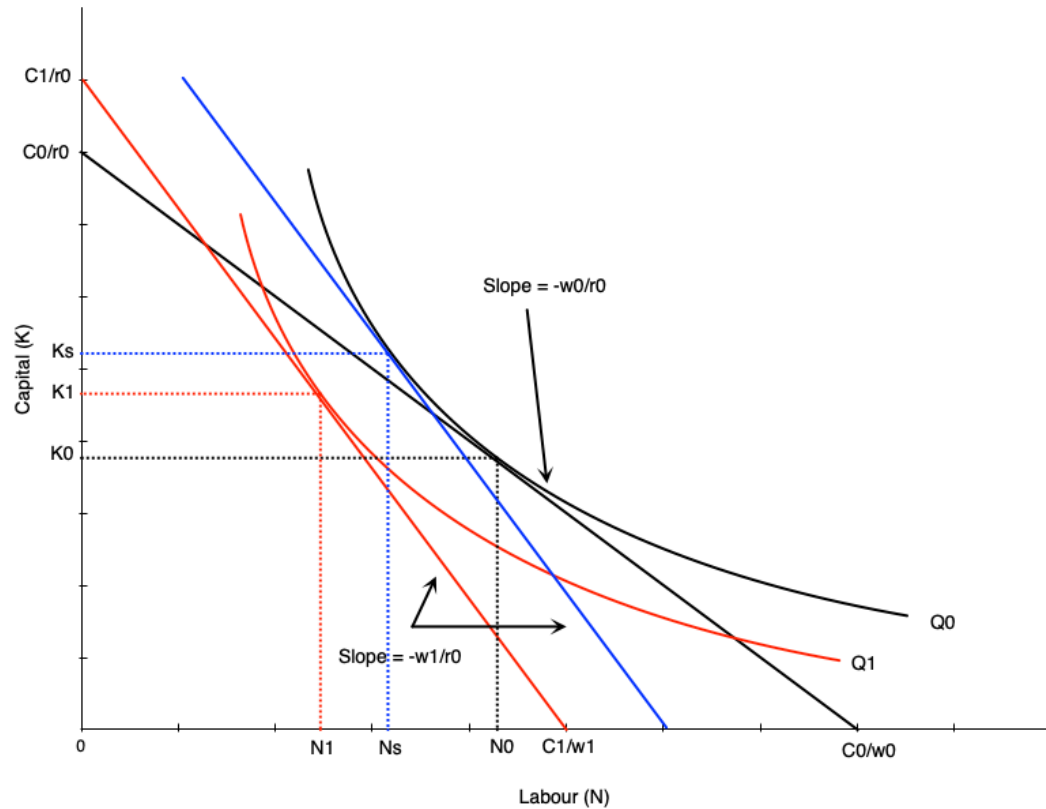
For Labour Demand

Scale and substitution effects work in the **same direction** — both reduce labour when wage rises

For Capital Demand

Scale and substitution effects work in **opposite directions** — effect is ambiguous

Scale and Substitution Effects



To find substitution effect:

- Find cost minimizing bundle at new wage and old output
- Blue line in graph
- Substitution effect: $N_0 \rightarrow N_s$

Scale effect:

- Change from optimum at new wage/old output to new wage/new output
- Shift from blue to red line in graph
- Scale effect: $N_s \rightarrow N_1$

Scale and Substitution Effects

Notice that scale and substitution effects have opposing effects on capital

When wage rises:

- **Substitution effect** → more capital used (firms substitute capital for labour)
- **Scale effect** → less capital used (firms reduce output, use less of all inputs)

 **Warning**

Effect on capital is **ambiguous**

Relationship to Short-Run Demand

In the short run:

- Capital is fixed → no substitution effect

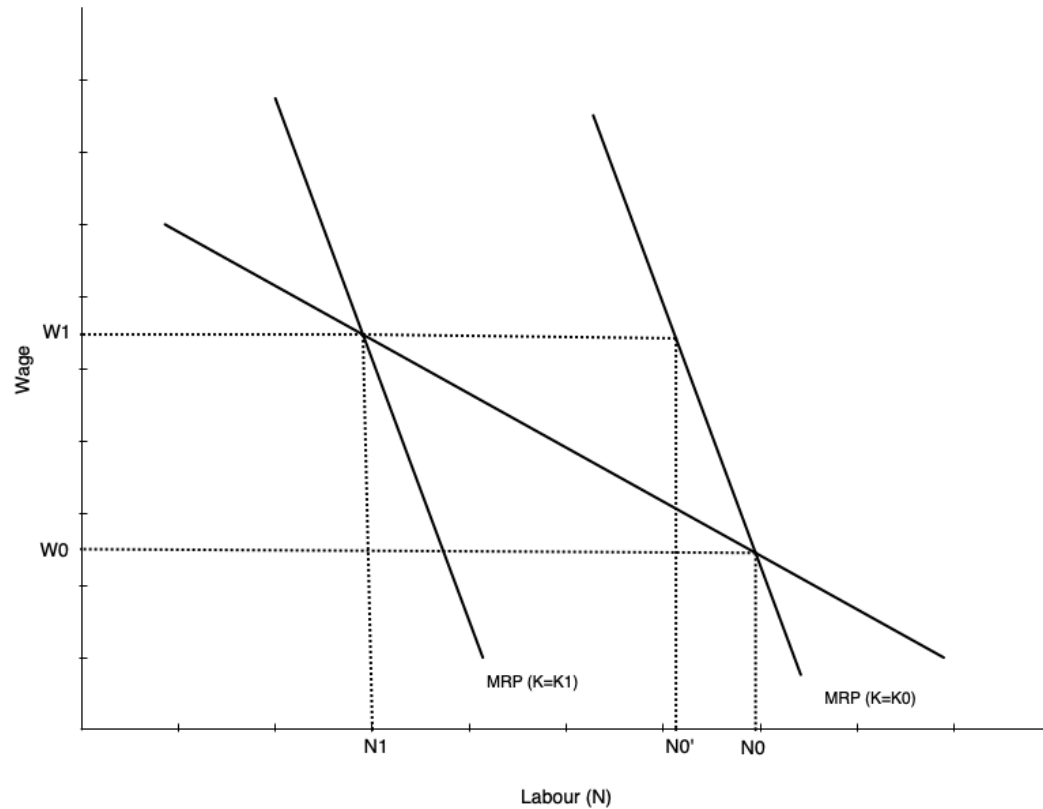
In the long run:

- Both substitution and scale effects occur

Key Result

Long-run labour demand is **more elastic** than short-run labour demand — firms can adjust all inputs in the long run, making them more responsive to wage changes

Relationship to Short-Run Demand



Graph shows short-run and long-run labour demand curves

- Start with capital fixed at K_0
- Wage goes from w_0 to w_1
- In short run: move along SR demand curve from N_0 to N_1
- In long run: firm substitutes capital for labour
- Short run curve shifts left
- Equilibrium at intersection of wage and new short run demand

Discussion

When wages rise, the substitution and scale effects both reduce labour demand, but they have opposing effects on capital demand. Can you think of a real-world industry where the substitution effect on capital would clearly dominate the scale effect, and one where the scale effect might dominate? What makes those industries different?

Elasticity of Demand for Labour

Elasticity of Demand

Recall: elasticity of demand measures responsiveness of quantity demanded to changes in price

In this case, responsiveness of labour demanded to changes in wage rate

Elastic labour demand curve:

- Proportionately larger change in quantity than change in wage
- Curve is flatter

Inelastic labour demand curve:

- Proportionately smaller change in quantity than change in wage
- Curve is steeper

Factors Affecting Elasticity

Several factors affect elasticity of labour demand:

1. Ease of substitution between labour and other inputs
 - More/easier substitutes → more elastic demand
2. Elasticity of supply of other inputs
 - Inelastic supply of other inputs → harder to substitute → less elastic labour demand
 - Happens because other inputs become more expensive when firm tries to substitute
3. Elasticity of demand for the final product
 - Output responding more to price changes → more elastic labour demand
4. Labour share in total costs
 - Higher labour cost share → more elastic labour demand (operates via scale effect)

Evidence on Elasticity

In the USA:

- Elasticity between **-0.15 and -0.75** (relatively inelastic)
- Elasticities have increased over time
- Long run elasticities are larger

In Canada:

- Elasticity between **-0.1 and -0.6**
- Depends on geography and industry

Key Result

Studies confirm that labour demand does slope down

Discussion

Why might we observe that labour demand has become more elastic over time in both Canada and the USA? Which of the four factors affecting elasticity do you think has changed the most, and why?

Adjusting Workers versus Hours Worked

Background

Previous Assumption

Labour is a purely variable input paid a wage per hour worked

In reality it is not so simple:

- Workers are sometimes paid other benefits besides wages
 - CPP, EI, retirement benefits, paid vacation, sick leave, etc.
- There are costs beyond the hourly wage
 - Hiring and training costs
 - Costs of dismissal
 - Non-wage benefits described above

These benefits and costs complicate the labour demand decision

Quasi-fixed Costs

Introduction

⚠ Important Distinction

Some costs are incurred per worker regardless of hours worked. Others are incurred per hour worked.

Quasi-fixed costs: costs that occur per worker but are relatively independent of hours worked

- Pure quasi-fixed costs do not vary with hours worked at all
- In practice, many costs are partially quasi-fixed
- Some recurring (e.g. monthly EI premiums), others one-time (e.g. hiring and training costs)

Examples of Quasi-fixed Costs

Hiring costs:

- Recruiting, interviewing, and training are costly
- Incurred when a worker is hired regardless of hours worked
- Firms generally try to minimize hiring costs by retaining workers

Termination costs:

- Severance pay, legal fees, and administrative costs
- Initially depend on time with company, but have a maximum so become fixed
- Firms may be reluctant to lay off workers because of these costs

Examples of Quasi-fixed Costs

Employer contributions to payroll taxes:

- EI and CPP are initially related to work hours
- But they have a ceiling so become **quasi-fixed** for high hours worked

Other non-wage benefits:

- Employer contributions to life insurance, medical, dental plans
- Often have a fixed component regardless of hours worked

Note

If a cost always varies with hours worked it is **not** quasi-fixed — e.g. hourly wage, overtime premiums, per-hour health and safety costs

Effects of Quasi-fixed Costs

Previously: labour demand determined by equating MRP_L to the wage rate

With quasi-fixed costs, the cost of hiring a worker is higher. There is also a **dynamic component** — firms amortize costs over time:

- Firms may pay high training costs now
- But expect the worker to remain to recoup those costs

A profit maximizing firm will hire workers up to the point where:

$$H + T + \sum_{t=1}^N \frac{w_t}{(1+r)^t} = \sum_{t=1}^N \frac{VMP_t}{(1+r)^t}$$

Where H = hiring cost, T = training cost, w_t = wage, VMP_t = value of marginal product, r = discount rate, N = expected tenure

Effects of Quasi-fixed Costs

This equation has several implications:

- Firms reluctant to hire workers for short periods if hiring costs are high
- Wage does not have to equal marginal product in each period
- Future productivity matters, but is discounted
- Firms may pay more than current marginal product if future productivity is expected to be high

When hiring costs are a factor, future productivity must exceed the total wage:

$$\sum_{t=1}^N \frac{VMP_t}{(1+r)^t} > \sum_{t=1}^N \frac{w_t}{(1+r)^t}$$

Effects of Quasi-fixed Costs

Like Any Other Investment Decision

Firms invest in workers when the present value of future returns exceeds the present value of costs

- Hiring and training costs → “upfront” investment costs
- Wages → “ongoing” investment costs
- Future productivity → returns on investment
- You only make the investment if you expect to earn at least as much as you spend

Effects of Quasi-fixed Costs

	Costs					
Employment Duration (N)	Fixed (H + T)	Variable ($\sum W_t$)	Total	Benefits ($\sum VMP_t$)	Total Profit	Yearly Profit
1 year	\$15,000	\$25,000	\$40,000	\$30,000	-\$10,000	-\$10,000
2 years	15,000	50,000	65,000	60,000	-5,000	-2,500
3 years	15,000	75,000	90,000	90,000	0	0
4 years	15,000	100,000	115,000	120,000	5,000	1,250
5 years	15,000	125,000	140,000	150,000	10,000	2,000

Think About It

What would happen to firm hiring behaviour if the government introduced a large new per-worker tax that does not vary with hours worked?

Things Explained by Quasi-Fixed Costs

Overtime

Firms sometimes work employees longer despite higher overtime wages

Explanation

High quasi-fixed costs of hiring additional workers make it cheaper to pay overtime than incur hiring and training costs — especially true for high-skill workers requiring a lot of training

Temp and Gig Workers

Firms do not always hire part- or full-time employees directly

They sometimes use temp agencies or “gig” workers:

- Laurier hires a cleaning firm and does not employ many janitors directly
- Uber classifies drivers as independent contractors

Explanation

By doing this, firms avoid paying quasi-fixed costs

Layoffs

During downturns firms sometimes keep workers around

Explanation: Labour Hoarding

- Paying severance and legal fees may be costly
- Firms may prefer to keep workers on payroll if they expect a quick recovery
- Firms reluctant to lay off skilled workers due to high training costs

This also explains why **highly educated workers** tend to suffer less unemployment during recessions

Layoffs

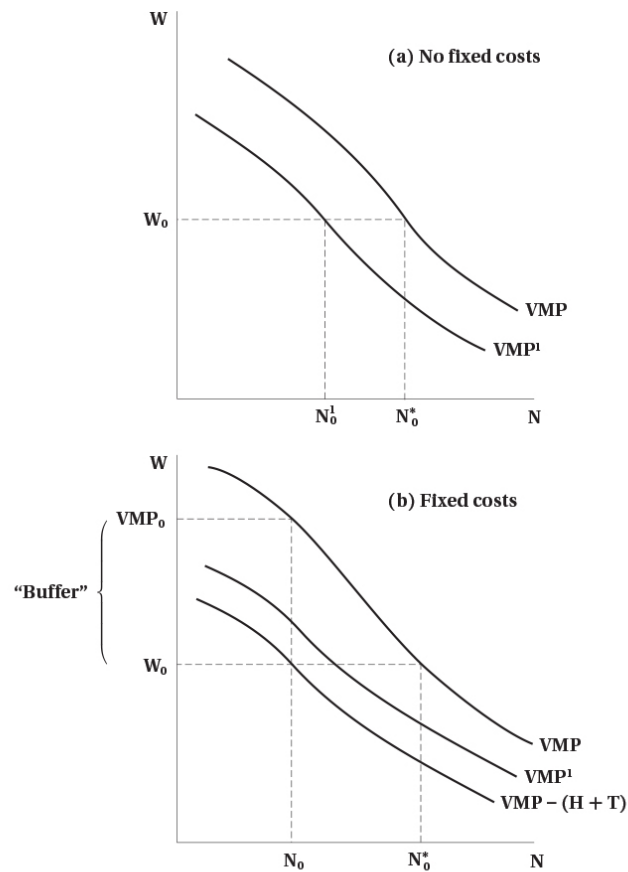


FIGURE 6.3 Nonrecurrent Fixed Employment Costs and Changes in Labour Demand

In panel (a), a decline in market conditions leads to an inward shift of the VMP schedule, from VMP to VMP¹. The profit-maximizing level of employment falls from N_0^* to N_0^1 at the prevailing wage of W_0 . By contrast, if there are fixed costs of hiring workers, employment may not fall as much. First, the original level of employment will be at N_0 , not N_0^* , as employers choose employment so that the wage equals the VMP net of hiring and training costs, $VMP - (H + T)$. At N_0 , the VMP (VMP_0) exceeds W_0 by a "buffer," $H + T$. As long as the new $VMP^1 > VMP - (H + T) = W_0$, it will not be optimal for the firm to change its employment level from N_0 .

Top panel — without quasi-fixed costs:

- Firm hires to point where $VMP = w$
- In downturn, VMP shifts down (lower demand for output)
- With same wage, firms hire fewer workers

Layoffs

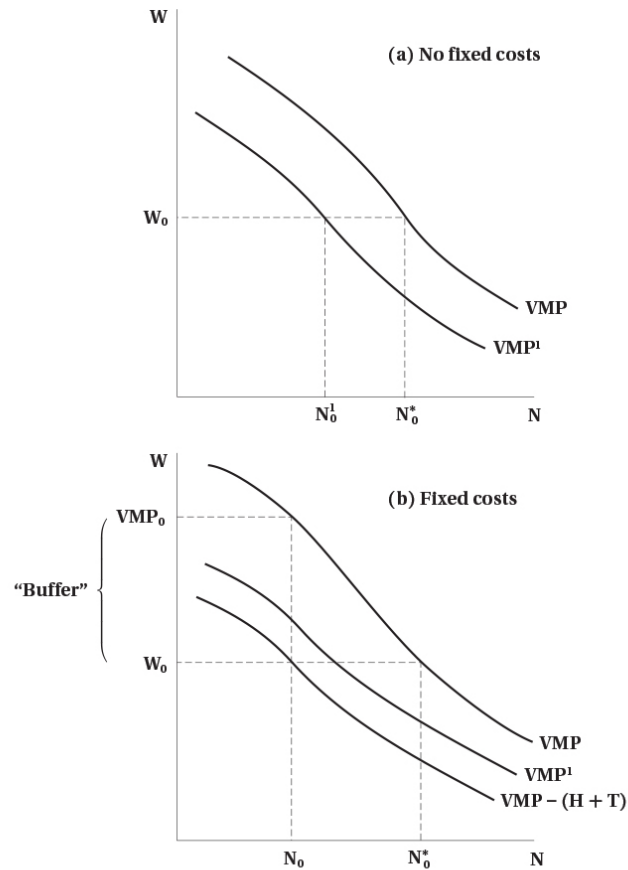


FIGURE 6.3 Nonrecurring Fixed Employment Costs and Changes in Labour Demand

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Bottom panel — with quasi-fixed costs:

- Quasi-fixed costs shift VMP curve down prior to downturn
 - Firm hires to point where $VMP - (H + T) = w$
- Creates wedge between VMP and wage
- After downturn, VMP shifts down
 - But because of wedge, firm may not change employment

Segmentation

Hiring costs mean trained and untrained employees are different

- Firms reluctant to hire untrained workers because of high training costs
- When quasi-fixed costs are high, firms prefer to retain or hire trained workers
- Leads to a **segmentation** into protected and unprotected workers

Also leads firms to work employees more intensively rather than hire new workers

Example

Big law firms work associates long hours instead of hiring new associates

Think About It

Why might we observe that highly skilled workers are more likely to work overtime, while lower-skilled positions are more likely to be filled by temp or gig workers?

Work Sharing

Some individuals would embrace work sharing

- Two or more people share a job
- Hire more people to each work less hours for same total hours

Why Firms Resist

Quasi-fixed costs make it costly to bring on additional workers — firms prefer fewer workers working more hours, even if total hours worked is the same

Work Sharing

Introduction

Canada's Work Sharing Program

Government program that subsidizes employee wages during downturns, administered through Employment Insurance

How it works:

- Allows firms to reduce hours instead of laying workers off
- Employees form a “work sharing group” and agree to reduce hours
- Maximum duration of **26 weeks** with possible extensions

The key idea is to avoid layoffs and employ more people

Introduction

Other possible work sharing schemes:

- Delayed entry into the labour force
- Early retirement
- Short work weeks
- Reduced overtime
- Increased vacation time

Key Question

Do any of these schemes actually increase the number of jobs?

Overtime Restrictions

One way to share work is to restrict overtime

- Some governments have enacted legislation to prevent excessive overtime
- Firms do not like this because it could increase costs

It is not clear that these laws increase employment:

- Firms may not hire more workers
- Increased costs reduce overall demand for labour
- New workers may not be as productive as existing workers
- There may not be enough employable workers available

Overtime Restrictions

Lump of Labour Fallacy

The belief that there is a fixed number of jobs in the economy — leading people to think employees working longer hours take jobs away from others

In reality:

- The number of jobs is not fixed
- As people become employed, the economy grows and more jobs are created
- Fallacy is commonly used to argue against immigration

Evidence

Studies show that restricting work hours does not lead to increased employment — but may still be useful in the short term during economic downturns

Subsidized Work Sharing

Canada subsidizes work sharing programs for economic downturns

- Main goal: keep people employed during recessions
- Would be good if it also protects jobs longer term

Evidence

- Many firms use the program
- It does help keep people employed during downturns
- But **little evidence** that it increases employment longer term
 - Employees were laid off when subsidy ran out

Employee Demand for Work Sharing

Key Point

Employees need to also buy in for work sharing to be successful — it is not clear that they do in many cases

Example: union jobs

- Unions often resist work sharing programs
- Prefer to keep full-time jobs with benefits
- May see work sharing as a threat to job security
- May also see it as a way for employers to reduce wages and benefits

Discussion

Given what we know about the lump of labour fallacy, why do politicians and the public still frequently argue that reducing work hours or restricting immigration will create more jobs?

Non-Wage Benefits

Non-wage Benefits

Compensation typically includes more than just wages

Common non-wage benefits:

- Health insurance (beyond public health care)
- Retirement benefits
- Paid vacation and sick leave
- Other perks (gym memberships, childcare, etc.)
- These benefits have value to workers
- Increased as a share of total compensation through the 90s and early 2000s

Non-wage Benefits

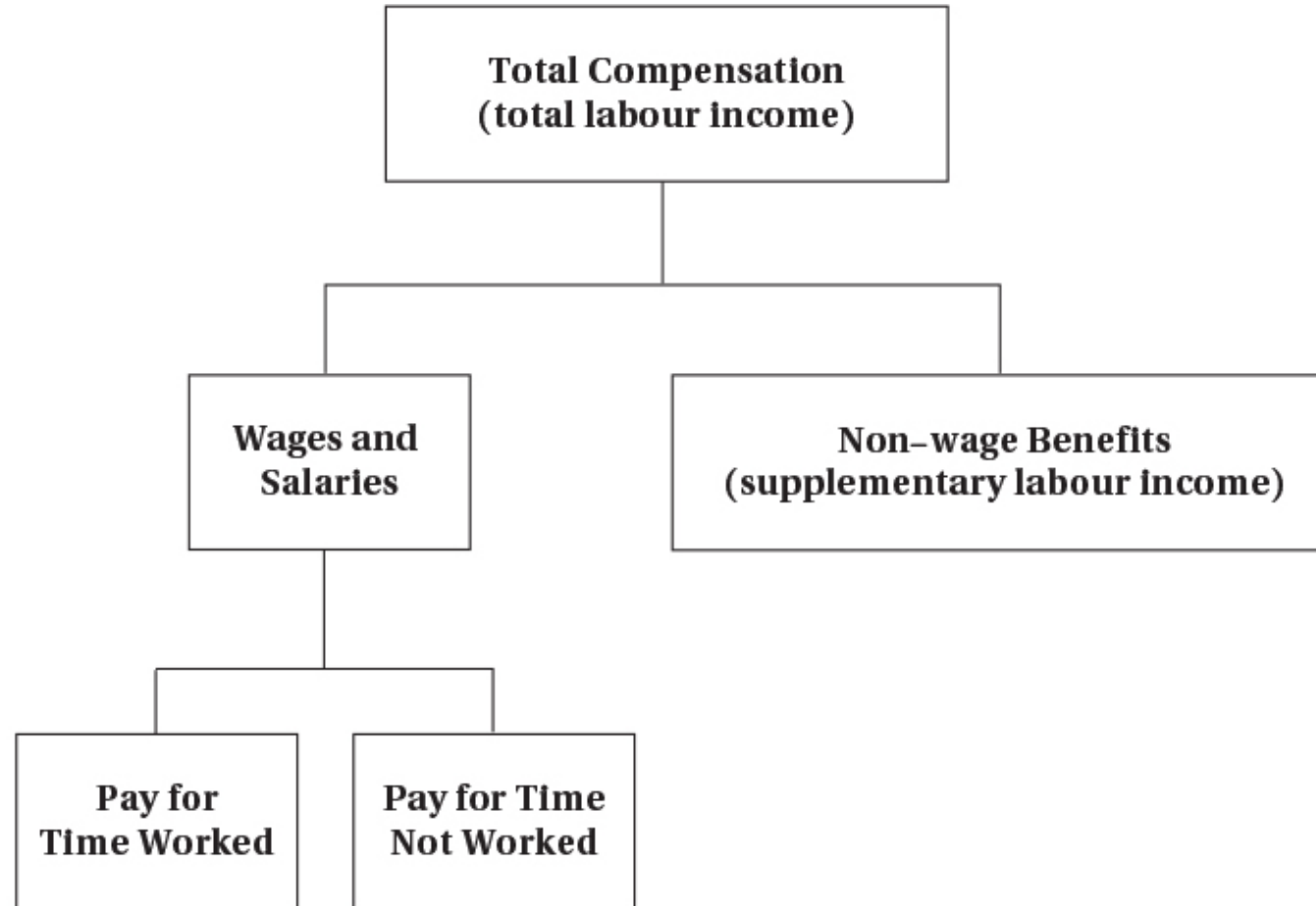


FIGURE 6.1 Components of Total Compensation

Total compensation includes wages plus non-wage benefits. The wage and salary component can be further divided into the part directly corresponding to time worked and that which does not depend on time worked (such as vacation pay).

Non-wage Benefits

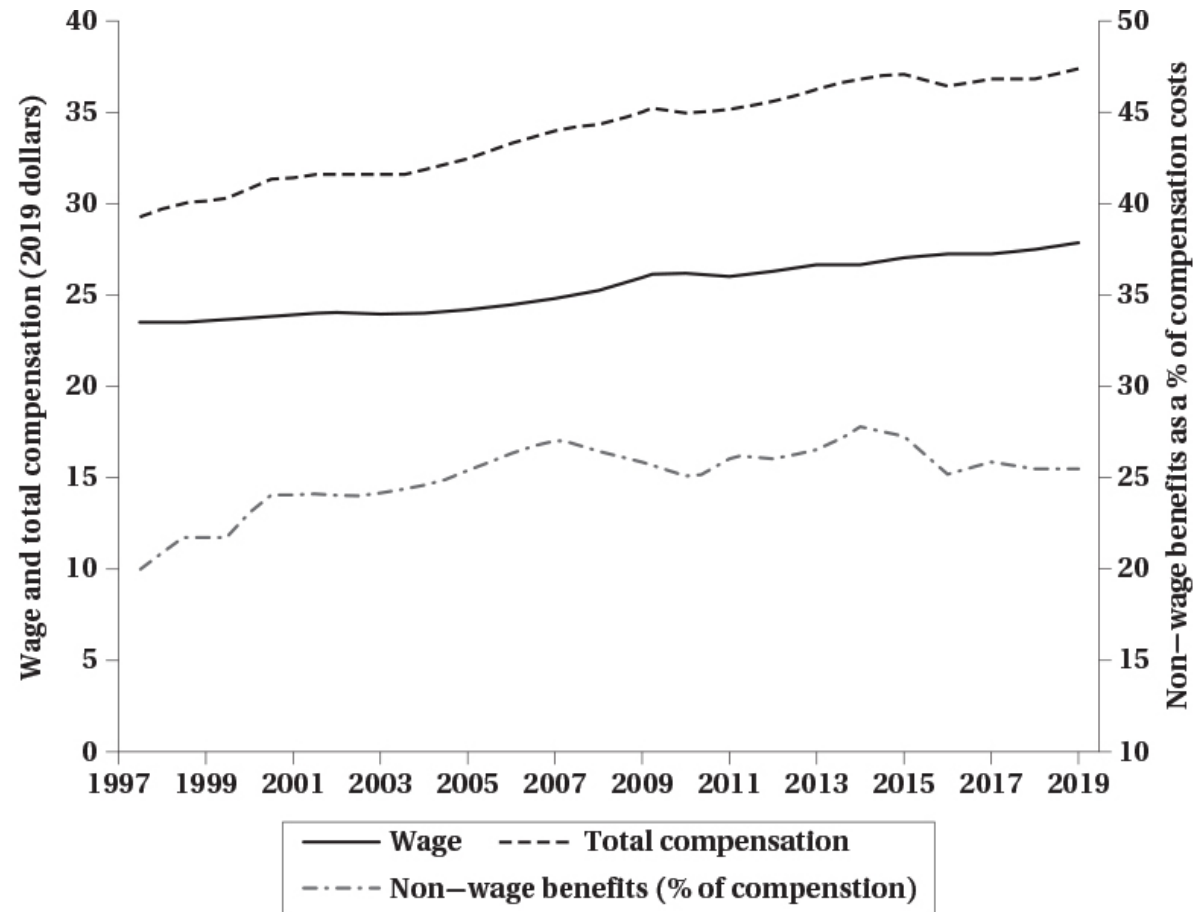


FIGURE 6.2 Total Compensation, Wage, and Non-wage Benefits

This figure shows the evolution of total compensation per hour, wages per hour, and non-wage benefits as fraction of total compensation. Total compensation and wages are expressed in dollars of 2019. Total compensation is growing faster than wages, reflecting the growing contribution of non-wage benefits to total compensation.

SOURCE: Statistics Canada table 36-10-0489-01 (total compensation from the System of National Accounts) and table 14-10-0064-01 (hourly wage from the Labour Force Survey).

Non-wage Benefits

Why do firms pay benefits instead of wages?

- Tax advantages: sometimes benefits are not taxed
- Economies of scale: pensions and insurance are cheaper when provided to a group (avoids adverse selection)
- Worker preferences: risk-averse workers may prefer health insurance to wages
- Legal requirements

Non-wage Benefits

More reasons:

- Better firm planning: benefits like sick leave reduce absenteeism and provide contingency plans
- Productivity: subsidized meals keep people working
- Deferred compensation: retirement benefits help retain workers
 - Workers may accept lower wages in exchange for future benefits
 - Acts as a screen for good workers

Non-wage Benefits

Governments also prefer and mandate them:

- Firm pensions reduce need for public pensions
- Better health outcomes reduce public health care costs
- Improve social safety net through sick leave and unemployment insurance



Tip

With all these reasons, it is not surprising that non-wage benefits are common

Discussion

If non-wage benefits are so advantageous for both firms and workers, why do some firms still offer minimal benefits and rely almost entirely on wage compensation?

Summary

Summary

Labour demand:

- In the short run firms hire to the point where the wage equals the value of the marginal product
- In the long run they minimize cost, and a wage change has scale and substitution effects
- Long-run demand is more elastic than short-run demand
- Elasticity depends on substitutes, the elasticity of output demand, and the labour cost share
- Quasi-fixed costs make workers and hours imperfect substitutes, and explain overtime, labour hoarding, and segmented workforces